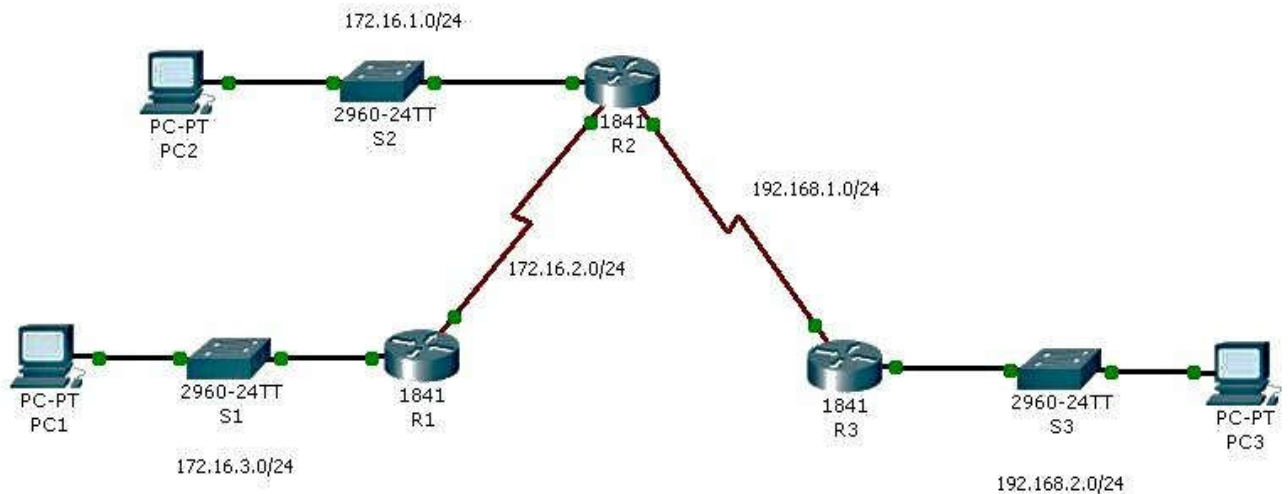


The Computer Networks

Lab 3

Static Routing

The purpose of this Lab is to implement a static routing configuration.



1. Review of the router connections

The goal is to create the topology that you will use throughout this lab. You add all the necessary peripherals and connect them with the appropriate wiring.

1.1: Create the topology

Task 1: physical installation and device configuration

Step 1: installing devices in the topology

Drag three 1841 routers from the device location on the logical workplace and drop them on labels R1, R2 and R3.

Drag three 2960 switches from the location of the device on the logical workplace and drop them on the labels S1, S2 and S3.

Drag three generic computers for the location of the device instead of logical work and drop them on labels PC1, PC2 and PC3.

Step 2: Adding modules to the routers. Proceed as follows for each of the three routers:

Click on the router. The router configuration window displays with the Physical tab. Turn the router off.

Drag a WIC-2T module to the open slot on the right.

Power on the router and click the CLI tab to start the process. After the startup, click the Setup tab.

Task 2: Naming the devices

Step 1: modification of the devices display names

Go to the router R1. The router configuration page displays, and presents the selected Config tab. Set the Display Name on R1.

Change the display names of the other two routers to suit the labels.

Change the display names on the three other switches, to match the labels.

Change the display names on the three other computers to match the labels. To avoid conflict of names, first change the PC3 display name and that of PC2 to finish with the PC1.

Task 3: Device Connection

Step 1: routers connecting to each other

Select the connections to the device location, and use a DCE serial cable to connect interface Serial0/0/0 of router R1 to serial0/0/0 of router R2. The first enabled router is connected to the DCE end of the cable. This is why it is recommended to start from the router R1.

Use a DCE serial cable to connect serial0/0/1 of router R2 to serial0/0/1 of router R3. The first enabled router is connected to the DCE end of the cable. This is why it is recommended to start with the router R2.

Step 2: Connect the local networks (LAN)

Use a straight copper cable to connect the interface FastEthernet0/0 on all three routers to the first port on the corresponding switch.

Use a straight copper cable to connect the interface FastEthernet on three computers at the next port of the corresponding switch.

2. Check the Ethernet interface

The goal is to apply the configuration of Ethernet interfaces. Follow the additional instructions to examine the ARP process in simulation mode.

Task 1: Configure IP information on Ethernet interfaces of a network

Configure the following IP information:

Peripheral	Interface	IP address	Subnet mask	Default Gateway
R1	Fa0/0	172.16.3.1	255.255.255.0	N/D
	S0/0/0	172.16.2.1	255.255.255.0	N/D
R2	Fa0/0	172.16.1.1	255.255.255.0	N/D
	S0/0/0	172.16.2.2	255.255.255.0	N/D
	S0/0/1	192.168.1.2	255.255.255.0	N/D
R3	Fa0/0	192.168.2.1	255.255.255.0	N/D
	S0/0/1	192.168.1.1	255.255.255.0	N/D
PC1	NIC	172.16.3.10	255.255.255.0	172.16.3.1
PC2	NIC	172.16.1.10	255.255.255.0	172.16.1.1
PC3	NIC	192.168.2.10	255.255.255.0	192.168.2.1

Task 2: verifying the proper operation

Step 1: Check the status of interfaces on routers. Check the status of FastEthernet interfaces by entering the “show ip interface brief” command on the three routers.

Step 2: Check the connectivity between hosts and routers. Enter the command “arp -a” from the PC1 prompt commands. Enter the “show arp” command on router R1. Note the results. Enter the “ping 172.16.3.1” command from prompt commands of PC1. Enter the command “arp -a” from the PC1 prompt commands. Enter the “show arp” command on router R1. Note the results. Each of the devices now has an input of the other device in its ARP table.

Step 3: Observe the ARP operation. Switch to Simulation mode. Set Event List Filters to display only the ARP and ICMP events. Launch the “ping 172.16.1.1” command from PC2 prompt commands. Reduce the window of the command prompt. Click the button Auto Capture / Play. Note that ARP packets are sent before the ICMP packets.